EXERCISE-I (Conceptual Questions) — Build Up Your Understanding

CLASSIFICATION

1. The hybrid state of C-atoms which are attached to a single bond with each other in the following structure are : $CH_2=CH-C\equiv CH$

(1) sp^2, sp (2) sp^3, sp
(3) sp^2, sp^2 (4) sp^2, sp^3

NC0001

2. The third member of the homologous series of aliphatic aldehydes has the structure :-

(1) CH_3CH_2CHO
(2) $CH_3(CH_2)_2CHO$
(3) $CH_3COCH_2CH_3$
(4) CH_3COCH_3

NC0002

3. Molecular formula $C_4H_8O_2$ represents :-

(1) An acid only
(2) An ester only
(3) An alcohol only
(4) An acid and an ester also

NC0003

4. The higher homologue of dimethylamine ($CH_3-NH-CH_3$) has the structure :-

(1) $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle |}{N}}-CH_3$
(2) $CH_3-CH_2-CH_2-NH_2$
(3) $CH_3-NH-CH_2-CH_3$
(4) $CH_3-\underset{\displaystyle NH_2}{\overset{\displaystyle |}{CH}}-CH_3$

NC0004

5. The third member of the family of alkenynes has the molecular formula :-

(1) C_6H_6 (2) C_5H_6
(3) C_6H_8 (4) C_4H_4

NC0005

6. The number of olefinic bonds in the given compound is/are :-

$$CH_2=CH-\underset{\displaystyle O}{\overset{\displaystyle \|}{C}}-CH=CH-C\equiv N$$

(1) 2 (2) 3
(3) 1 (4) 4

NC0006

7. The number of acetylinic bonds in the given compound is/are :

$$HC\equiv C-\underset{\displaystyle O}{\overset{\displaystyle \|}{C}}-CH=CH-C\equiv N$$

(1) 2 (2) 3 (3) 1 (4) 4

NC0007

8. The number of C-atoms in second member of an ester is/are :

(1) 2 (2) 3 (3) 4 (4) 5

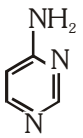
NC0008

9. Which of the following is an example of symmetrical or simple ether :

(1) $CH_3-\underset{\displaystyle O}{\overset{\displaystyle \|}{C}}-CH_3$
(2) $CH_3-O-CH_2-CH_3$
(3) $CH_3-CH_2-O-CH_2-CH_2-CH_3$
(4) $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle |}{CH}}-O-\underset{\displaystyle CH_3}{\overset{\displaystyle |}{CH}}-CH_3$

NC0009

10. The number of hetero atoms present in the following compound is/are :



(1) 2 (2) 3 (3) 1 (4) 4

NC0010

11. Which of the following compound has sp-hybridised carbon atom :-

(1) CH_3COOH
(2) CH_3COCH_3
(3) CH_3CH_2CN
(4) $CH_2=CH-CH=CH_2$

NC0012

12. In compound $HC\equiv C-CH_2-CH=CH-CH_3$, the C_2-C_3 bond is the type of :-

(1) $sp - sp^2$
(2) $sp^3 - sp^3$
(3) $sp - sp^3$
(4) $sp^2 - sp^2$

NC0013

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13. Which of the following pair of compounds are homologues :-
(1) 1–Propanol & 2–Propanol
(2) Ethanol & Propanal
(3) Acetone & Acetaldehyde
(4) Acetic acid & Butyric acid

NC0015

14. Which of the following homologous series has incorrect general formula :-

(1) Alkyne	C_nH_{2n-2}
(2) Alkanol	$C_nH_{2n+2}O$
(3) Alkanal	$C_nH_{2n+1}O$
(4) Carboxylic acid	$C_nH_{2n}O_2$

NC0016

15. Minimum number of carbon atoms present in an ester are :-
(1) 2 (2) 1 (3) 4 (4) 3

NC0018

16. Which of the following has general formula C_nH_{2n}
(1) Only Alkyne
(2) Only Alkane
(3) Aromatic hydrocarbon
(4) Alkene & cyclic Alkane

NC0019

COMMON AND DERIVED NAME

17. Which of the following are tertiary radicals :-

(a) $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}-$ (b) $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle H}{\overset{|}{\underset{|}{C}}}}-$

(c) $CH_3-\underset{\displaystyle C_2H_5}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}-$ (d) $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}-CH_2-$

(1) a and b (2) b and c
(3) a and c (4) b and d

NC0021

18. Common name of the given compound is :-

$$CH_3-\overset{\displaystyle O}{\overset{\|}{C}}-O-CH=CH_2$$

(1) vinyl acetate
(2) acryl acetate
(3) methyl acrylate
(4) Vinyl ethanoate

NC0022

19. A primary amine has amino group ($-NH_2$) attached to:-
(1) A primary carbon atom only
(2) A secondary carbon atom only
(3) A tertiary carbon atom only
(4) A primary, secondary or tertiary carbon atom

NC0023

20. Which of the following are secondary radicals :-

(a) $CH_3-\overset{\displaystyle H}{\overset{|}{C}}H-C_2H_5$ (b) $CH_2=\overset{\displaystyle H}{\overset{|}{C}}-CH_3$

(c) $CH_2=CH-$ (d) $(CH_3)_2CH-$

(1) a, b, c (2) a, d, c
(3) b, c, d (4) a, b, d

NC0024

21. Examine the following structures :-

(A) $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}-OH$; (B) $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}-NH_2$

Which of the following statement is correct :-
(1) A is tertiary alcohol while B is tertiary amine
(2) A is primary alcohol while B is primary amine
(3) A is tertiary alcohol while B is primary amine
(4) A is primary alcohol while B is tertiary amine

NC0025

22. Which of the following is not a correct match

(1) $H_3C-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}-CH_2-$ ⇒ Neopentyl

(2) $H_3C-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}-$ ⇒ Neobutyl

(3) $HC\equiv C-CH_2-$ ⇒ Propargyl

(4) $CH_2=CH-CH_2-$ ⇒ Allyl

NC0026

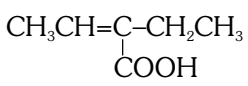
IUPAC NAME

23. The IUPAC name for isobutyl chloride is :-
(1) 2–Methyl-2–chloro butane
(2) 2–Chloro–2–methyl butane
(3) 1–Chloro–2–methyl propane
(4) 2–Methyl–3–chloro propane

NC0027

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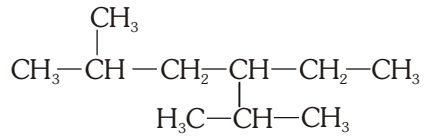
24. The IUPAC name of given compound is :-



(1) 3–Carboxy–2–pentene
(2) 2–Ethylidene butanoic acid
(3) 2–Ethyl–2–butenoic acid
(4) 3–Ethyl–2–buten–4–oic acid

NC0028

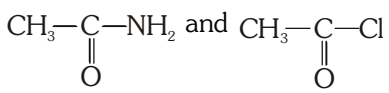
25. The IUPAC name for the given structure is :-



(1) 3–Isopropyl–4–methylhexane
(2) 4–Isopropyl–3–methylhexane
(3) 3–Ethyl–2,5–dimethylhexane
(4) 2–Ethyl–3–isopropylpentane

NC0029

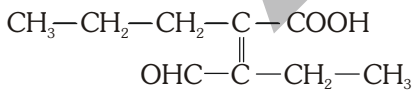
26. The IUPAC name for



(1) 1–Amino–1–oxo ethane, 1–chloro ethanal
(2) 1–Amino ethanal, acetoyl chloride
(3) 1–Oxoethanamine, ethanoyl chloride
(4) Ethanamide, Ethanoyl chloride

NC0030

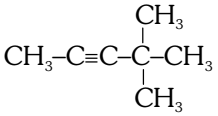
27. The number of carbon atoms in the principle chain of the given compound are :-



(1) 7 (2) 5 (3) 4 (4) 6

NC0031

28. The IUPAC name of given compound is :-



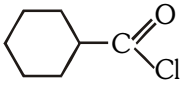
(1) Methyl tertiarybutyl acetylene
(2) t–Butyl propyne
(3) 4,4–Dimethyl–2–pentyne
(4) 1,3,3,3–Tetramethyl ethyne

NC0032

29. The IUPAC name of $CH_3—CH_2—NH—CH_3$ is :-
(1) Methyl ethyl amine
(2) 1–methyl amino ethane
(3) N–methyl ethan amine
(4) N–ethyl methan amine

NC0034

30. The IUPAC name for the compound is :-



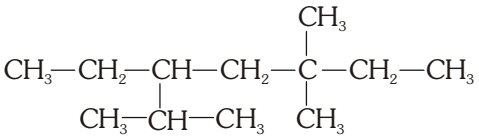
(1) Cyclohexanoyl chloride
(2) Cyclohexane carbonyl chloride
(3) 1–Chloro cyclohexanal
(4) Chloro cyclohexyl methanal

NC0035

31. The IUPAC name of $HC\equiv C-\underset{\underset{CH_3}{|}}{C}=CH-CH_3$ is
(1) 3–Methyl–2–penten–4–yne
(2) 3–Methyl–3–penten–1–yne
(3) 3–Methyl–4–pentyn–1–ene
(4) 3–Methyl pentenyne

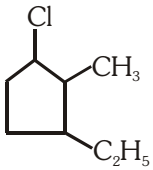
NC0036

32. The IUPAC name of the structure is :-



(1) 3–Isopropyl–5,5–dimethyl heptane
(2) 5–Ethyl–3,3,6–trimethyl heptane
(3) 3,3–Dimethyl–5–isopropyl heptane
(4) 3–Ethyl–2,5,5–trimethyl heptane

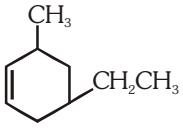
NC0037

33.  has the IUPAC name :-

(1) 3–Chloro–1–ethyl–2–methyl cyclopentane
(2) 1–Chloro–3–ethyl–2–methyl cyclopentane
(3) 4–Chloro–1–ethyl–5–methyl cyclopentane
(4) All are correct

NC0038

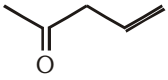
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34. The IUPAC name of  is :-

(1) 1–Methyl–5–ethyl cyclohex–2–ene
(2) 5–Ethyl–3–methyl cyclohex–1–ene
(3) 4–Ethyl–6–methyl cyclohex–1–ene
(4) 1–Ethyl–5–methyl cyclohex–3–ene

NC0039

35. The IUPAC name for the compound is :-

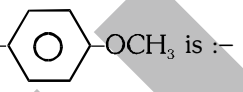


(1) 2-Acetyl prop-1-ene (2) Pent-1-en-4-one
(3) Pent-4-en-2-one (4) Formyl propene

NC0041

36. Which is incorrect IUPAC name :-
(1) 3–Pentyne
(2) 3–Methyl–2–butanone
(3) 2–Ethyl–3–methyl–1–butene
(4) 3–Ethyl-2-methyl pentane

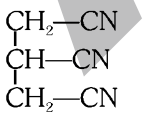
NC0042

37. The IUPAC name of  is :–

(1) 1–Methoxy–4–amino benzene
(2) Aminophenyl methyl ether
(3) 4–Methoxy aniline
(4) None of the above

NC0043

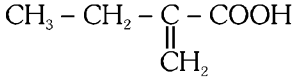
38. The IUPAC name of the given compound is :–



(1) 1,2,3–Tricarbonitrile propane
(2) Propane–1,1,1–tricarbylamine
(3) Propane–1,2,3–tricarbonitrile
(4) 3–Cyano pentane dicyanide

NC0045

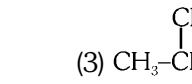
39. Number of carbon atoms in the principle carbon chain in the given compound are :-

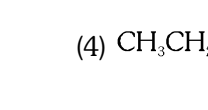


(1) 4 (2) 3
(3) 2 (4) 5

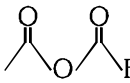
NC0046

40. Wrong IUPAC name is :-
(1) $CH_3CH_2CONH_2$ Propanamide
(2) $CH_3CH_2COOCH_3$ Methyl propanoate


(3) 2–Methyl pent–3– ene


(4) 2–Ethoxy butane

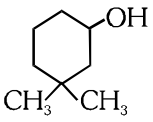
NC0047

41. The IUPAC name of  is :–

(1) Acetic anhydride
(2) Formyl ethanoate
(3) Butane– 2, 4–dione
(4) Ethanoic methanoic anhydride

NC0049

42. The IUPAC name of given compound is :



(1) 3,3–Dimethyl–1–hydroxy cyclohexane
(2) 1,1–Dimethyl–3–hydroxy cyclohexane
(3) 3,3–Dimethyl–1–cyclohexanol
(4) 1,1–Dimethyl–3–cyclohexanol

NC0050

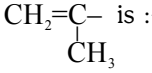
43. IUPAC name of $(CH_3)_2CHCH(CH_3)_2$ is :-
(1) 2,2–Dimethyl butane (2) 2,3–Dimethyl butane
(3) 2,4–Dimethyl butane (4) 1–Methyl pentane

NC0051

44. IUPAC name of CH_2═CH—CH_2—Cl is :–
(1) Allyl chloride
(2) 1–Chloro–3–propene
(3) 3–Chloro–1–propene
(4) Vinyl chloride

NC0052

45. The IUPAC name of the following group



(1) Isopropenyl
(2) 1–Methylethenyl
(3) 2–Methylethylnyl
(4) None of the above

NC0053

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46. $CH_3—CH=CH—C\equiv CH$ has IUPAC name :-
(1) Pent–2–en–4–yne
(2) Pent–4–yn–2–ene
(3) Pent–1–yn–3–ene
(4) Pent–3–en–1–yne

NC0054

47. The IUPAC name of the following compound

$CH_3CH_2–\underset{\displaystyle CH_3CH_2–CH–CH_2–CH_3}{\overset{\displaystyle |}{CH}}–CH_2–CH_3$

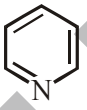
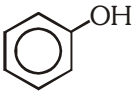
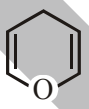
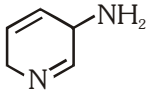
(1) 3, 4 – Dimethyl octane
(2) 3–sec pentyl pentane
(3) 3, 4 - Diethyl hexane
(4) 3, 4 – Dimethyl hexane

NC0055

48. Correct IUPAC name is :-
(1) 3-Methyl-2- ethylpentane
(2) 2-Ethyl- 3-methylpentane
(3) 3–Ethyl– 2-methylpentane
(4) 2-Ethyl- 2-methylpentane

NC0056

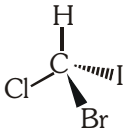
49. Which is hetero alicyclic ?

(a) 
(b) 
(c) 
(d) 
(e) 
(f) 

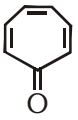
Correct choice is :
(1) a ,b, d and e
(2) c and e
(3) d and e
(4) only f

NC0112

50. Incorrect statement is :

(1)  Iodine atom is away from observer and –Cl is on the plane of paper

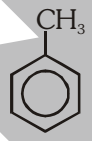
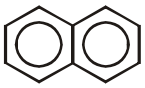
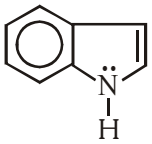
(2) Alicyclic are those compounds which have similar properties of aliphatic compounds

(3)  is benzenoid structure

(4) Naphthalene is example of fused benzene ring

NC0113

51. Which of the following is non-benzenoids ?

(1) 
(2) 
(3) 
(4) 

NC0114

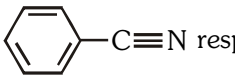
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EXERCISE-I (Conceptual Questions) — ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**1**	**1**	**4**	**3**	**3**	**1**	**3**	**2**	**4**	**1**	**3**	**3**	**4**	**3**	**1**
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	**4**	**3**	**1**	**4**	**4**	**3**	**2**	**3**	**3**	**3**	**4**	**2**	**3**	**3**	**2**
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	**2**	**4**	**2**	**2**	**3**	**1**	**3**	**3**	**2**	**3**	**4**	**3**	**2**	**3**	**2**
Que.	46	47	48	49	50	51									
Ans.	**4**	**3**	**3**	**3**	**3**	**3**									

EXERCISE-III (Analytical Questions)

Master Your Understanding

1. The number of π bonds and σ bonds in the structure  respectively are :-

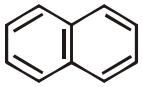
(1) 8, 5 (2) 8, 6
(3) 3, 8 (4) 5, 13

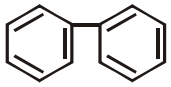
NC0078

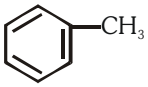
2. The geometry of the compound is :-
$CH_2=CH-CH=CH_2$
(1) Tetrahedral (2) Linear
(3) Planar (4) Pyramidal

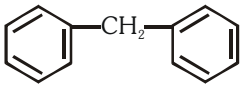
NC0079

3. Which of the following have only 2° H-atom :

(a) 

(b) 

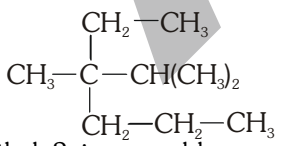
(c) 

(d) 

Correct code is :
(1) a and b (2) a, b and d
(3) a, c and d (4) a, b, c and d

NC0080

4. The IUPAC name of the given structure is :-



(1) 3–Methyl–3–isopropyl hexane
(2) 3–Isopropyl–3–methyl hexane
(3) 3–Ethyl–2,3–dimethyl hexane
(4) 2,3–Dimethyl–3–ethyl hexane

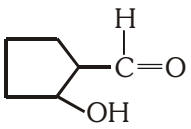
NC0081

5. The IUPAC name from the incorrect name 4–Amino–3–hydroxy–2–butene is :-
(1) 1–Amino–2–hydroxy–2–butene
(2) 4–Amino–2–buten–3–ol
(3) 1–Amino–2–buten–2–ol
(4) 1–Amino–2–butenol

NC0084

6. The correct IUPAC name of 2–chloro–3–butanol is :-
(1) 3–Chloro–2–hydroxy butane
(2) 3–Chloro–2–butanol
(3) 3–Hydroxy–2–chloro butane
(4) 2–Chloro–3–hydroxy butane

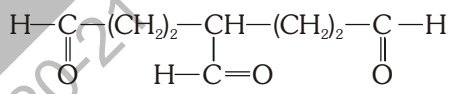
NC0085

7. The IUPAC name for  is :-

(1) 2–Hydroxy cyclopentanal
(2) 2–Formyl–1–hydroxy cyclopentane
(3) 2–Hydroxy cyclopentane carbaldehyde
(4) Cyclopentane–2–ol–1–al

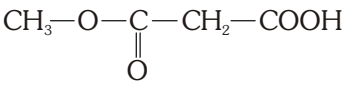
NC0086

8. The IUPAC name of the compound is :-



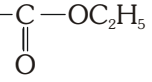
(1) 1,4,7–Triheptanal
(2) 1,3,5–Triformyl pentane
(3) 3–Formyl heptane dial
(4) 1,3,5–Pentane tricarbaldehyde

NC0087

9. 

The IUPAC name of the above compound is :-
(1) 2–Acetoxy ethanoic acid
(2) 2–Methoxycarbonyl ethanoic acid
(3) 3–Methoxyformyl ethanoic acid
(4) 2–Methoxyformyl acetic acid

NC0088

10. The IUPAC name of  is :-

(1) Ethoxy formyl chloride
(2) Ethoxy methanoyl chloride
(3) Ethyl chloro methanoate
(4) Ethoxy carbonyl chloride

NC0089

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11. The IUPAC name of CH_3CH_2NHCHO is :

(1) N–formyl ethanamine

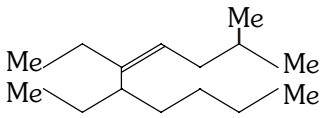
(2) Ethyl amino methanal

(3) N–ethyl methanamide

(4) N-ethyl methanol

NC0091

12. The IUPAC name of the structure is :-



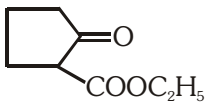
(1) 2,4,5–Triethyl–3–nonene

(2) 5,6–Diethyl–2–methyl–4–decene

(3) 2,4,5–Triethyl–3–octene

(4) 3–Ethyl–5–methyl–3–heptene

NC0092

13. 

has the IUPAC name :

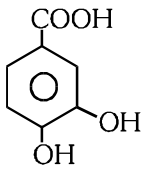
(1) Ethyl–2–keto cyclopentane carboxylate

(2) 2–Cyclopentanone–1–carbethoxy

(3) 2–Ethylcarbonate cyclopentanone

(4) 1–Keto–2–carbethoxy cyclopentanone

NC0093

14. The IUPAC name of 

is :-

(1) 3,4–Dihydroxy benzoic acid

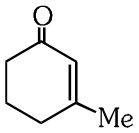
(2) 2,3–Dihydroxy benzoic acid

(3) 4–Carboxy–2–hydroxy phenol

(4) 4–Carboxy benzene–1,2–diol

NC0096

15. The IUPAC name of the given compound is :–



(1) 3–Methyl–2–cyclohexenone

(2) 2–Methyl–3–cyclohexenone

(3) 1–Oxo–3–methyl cyclohexene

(4) 2–Oxo–6–methyl cyclohexene

NC0097

16. The IUPAC name of 

is :–

(1) 2,3–Dimethyl hexane

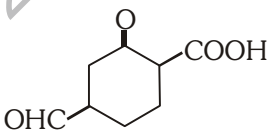
(2) 2–Ethyl–4–methyl pentane

(3) 3–Ethyl –2–methyl pentane

(4) 2,4–Dimethyl hexane

NC0098

17. IUPAC name of



is :-

(1) 6–Oxo–4–formyl cyclohexane carboxylic acid

(2) 4–Formyl–2–oxo cyclohexane carboxylic acid

(3) 4–Formyl–2–oxo cyclohexanoic acid

(4) 1–Carboxy–4–formyl–2–oxo cyclohexane

NC0099

18. Which of the following compounds has wrong IUPAC name ?

(1) $CH_3CH_2CH_2COOCH_2CH_3 \rightarrow$ Ethylbutanoate

(2) $CH_3–\underset{\displaystyle CH_3}{\underset{|}{CH}}–CH_2–CHO \rightarrow$ 3–Methylbutanal

(3) $CH_3–\underset{\displaystyle OH}{\underset{|}{CH}}–\underset{\displaystyle CH_3}{\underset{|}{CH}}–CH_3 \rightarrow$ 2–Methyl–3–butanol

(4) $CH_3–\underset{\displaystyle CH_3}{\underset{|}{CH}}–\overset{\displaystyle O}{\overset{\|}{C}}–CH_2–CH_3 \rightarrow$ 2–Methyl–3–Pentanone

NC0101

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19. Find correct IUPAC name of tetra-tert-butylmethane

(1) 3,3-di butyl-2,2,4,4-tetramethyl pentane

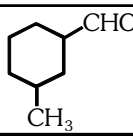
(2) 3,3-di tert-butyl-2,2,3,3-tetramethyl pentane

(3) 3,3-di tert-butyl-2,2,4,4-tetramethyl pentane

(4) 3,3-tert butyl-2,2,4,4-tetramethyl pentane

NC0116

20. Match the following structure given in column-I with their common names given in column-II and choose the correct option from the codes given below.

Column-I		Column-II	
A	$CH_3-\underset{\underset{Br}{|}}{CH}-CH_2-\overset{\overset{O}{||}}{C}-H$	X	Isobutyraldehyde
B	$(CH_3)_2CHCHO$	Y	α-Methoxy propionaldehyde
C		Z	β-Bromobutyraldehyde
D	$CH_3CH(OCH_3)CHO$	W	γ-Methyl cyclohexane-carbaldehyde

	A	B	C	D
(1)	X	Y	Z	W
(2)	Z	X	Y	W
(3)	Z	Y	X	W
(4)	Z	X	W	Y

NC0117

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EXERCISE-III (Analytical Questions) ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**4**	**3**	**2**	**3**	**3**	**2**	**3**	**4**	**2**	**3**	**3**	**2**	**1**	**1**	**1**
Que.	16	17	18	19	20										
Ans.	**4**	**2**	**3**	**3**	**4**										